Primer: Adversarial Inverse Reinforcement Learning (AIRL)

Fu, Luo, and Levine (2018)

econirl package documentation

1 Context and Problem

Maximum entropy IRL (Ziebart, 2010) recovers a reward function that rationalizes observed behavior, but the recovered reward entangles the true reward with the value function. For any bounded $h : \mathcal{S} \to \mathbb{R}$, the shaped reward

$$r'(s, a) \;=\; r(s, a) \;+\; \beta\, \mathbb{E}_{s' \sim P(\cdot|s,a)}[h(s')] \;-\; h(s) \tag{1}$$

produces the same optimal policy under the original transitions P (Ng et al., 1999). If P changes (new environment, counterfactual), the shaping term evaluates differently and the shaped reward no longer rationalizes the same behavior. Fu et al. (2018) propose AIRL to disentangle the transferable reward from the non-transferable value function by structuring the discriminator so that one component converges to $r^*(s)$ and the other absorbs $V^*(s)$.

2 Model and Notation

Table 1: Common DDC notation shared across all primers.

Symbol	Name	Definition
$s \in \mathcal{S}$	State	Discrete or continuous
$a \in \mathcal{A}$	Action	Finite: $\lvert\mathcal{A}\rvert = J$
s'	Next state	Drawn from $P(s' \mid s, a)$
$\beta \in [0, 1)$	Discount factor	
$\sigma > 0$	Scale parameter	Logit noise scale ($= 1$ by convention)
$\theta \in \Theta \subset \mathbb{R}^{d_\theta}$	Structural parameters	Unknown
$\phi(s, a)$	Feature vector	Known functions of (s, a), dimension K
$u(s, a; \theta)$	Flow utility	$\theta^\top \phi(s, a)$ (linear-in-parameters)
$P(s' \mid s, a)$	Transition kernel	Known or estimated from data
$Q(s, a)$	Choice-specific value	$u(s, a; \theta) + \beta \sum_{s'} P(s'\lvert s, a)\, V(s')$
$V(s)$	Integrated value	$\sigma \log \sum_a \exp(Q(s, a)/\sigma)$
$\pi(a \mid s)$	Choice probability	$\exp\big((Q(s, a) - V(s))/\sigma\big)$
τ	Trajectory	$(s_0, a_0, s_1, a_1, \ldots)$ from one agent
$\mathcal{D}$	Panel data	Collection of N observed trajectories

The agent maximizes entropy-regularized cumulative reward (Ziebart, 2010):

$$\max_{\pi}\; \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \beta^t \big(r(s_t, a_t) + \mathcal{H}(\pi(\cdot \mid s_t)) \big) \right]. \tag{2}$$

The optimal policy satisfies $\pi^*(a \mid s) = \exp(A^*)/\sum_{a'} \exp(A^*_{a'})$ where $A^*(s, a) = Q^*(s, a) - V^*(s)$, giving $\log \pi^*(a \mid s) = A^*(s, a)$.

Symbol	Meaning
$r^*(s)$	True reward (state-only; required for Theorem 1)
$\pi^*(a\|s)$, V^*, Q^*, A^*	Expert policy, value, action-value, advantage
$g_\theta(s)$	Reward approximator (state-only, param. θ)
$h_\phi(s)$	Shaping potential $\approx V(s)$ (param. ϕ)
$\pi_\omega(a\|s)$	Learned policy (param. ω)
$f_{\theta,\phi}(s,a,s')$	Discriminator logit (3); $D_{\theta,\phi} \in [0,1]$: discriminator output (4)

AIRL decomposes the discriminator into reward and shaping:

$$f_{\theta,\phi}(s,a,s') = g_\theta(s) + \beta\, h_\phi(s') - h_\phi(s), \tag{3}$$

$$D_{\theta,\phi}(s,a,s') = \frac{\exp(f_{\theta,\phi})}{\exp(f_{\theta,\phi}) + \pi_\omega(a \mid s)}. \tag{4}$$

The logit f mirrors the Bellman relationship $Q = r + \beta\mathbb{E}[V(s')]$, decomposed into $g_\theta \approx r$ and $h_\phi \approx V$. Note g_θ depends on s only, not on a.

3 Identification

Reward ambiguity. IRL rewards are identified only up to potential-based shaping (1) (Ng et al., 1999). The observed policy constrains $|\mathcal{S}|(|\mathcal{A}|-1)$ values but the reward has $|\mathcal{S}|\cdot|\mathcal{A}|$ unknowns, leaving $|\mathcal{S}|$ free variables corresponding to the value function (Cao and Cohen, 2021).

Decomposability (Definition B.1). An MDP is *decomposable* if all states are linked: for every pair (s,s') there exists a policy and time t such that s' is reachable from s with positive probability. This rules out absorbing states.

Disentanglement (Theorem 1). If (a) g_θ is state-only, (b) the MDP is decomposable, and (c) the adversarial training converges to the Nash equilibrium, then $g_\theta(s) = r^*(s) + c$ for a constant c.

Proof sketch. At Nash equilibrium $D = 1/2$, so $f = \log \pi^* = A^*$. Substituting (3): $g_\theta(s) + \beta h_\phi(s') - h_\phi(s) = r^*(s) + \beta V^*(s') - V^*(s)$. Since g_θ is state-only, taking the expectation over s' yields $g_\theta(s) = r^*(s) + h_\phi(s) - V^*(s) + c_0$. Decomposability implies $h_\phi(s) - V^*(s)$ is constant, so $g_\theta(s) = r^*(s) + c$. □

Limitations. Theorem 1 requires state-only rewards and excludes absorbing states. DDC models have action-dependent payoffs and exit options. Lee et al. (2026) resolve both with anchor constraints (see AAIRL primer).

4 Estimation Algorithm

Algorithm 1 AIRL (Fu et al. 2018, Algorithm 1)

Require: Expert demonstrations $\mathcal{D}_E = \{\tau_1, \ldots, \tau_N\}$, discount β
Ensure: Disentangled reward $g_\theta(s)$

```
Initialize θ_0, φ_0, ω_0
for iteration = 1, 2, ... do
    D_π ← roll out π_ω to collect M transitions                    // sample from policy
    for j = 1, ..., J do                                            // train discriminator
        f_n ← g_θ(s_n) + β h_φ(s'_n) − h_φ(s_n)                     // Eq. (3)
        D_n ← σ(f_n − log π_ω(a_n | s_n))                           // Eq. (4)
        L ← −E_{D_E}[log D] − E_{D_π}[log(1 − D)]
        θ, φ ← θ, φ − α_D ∇_{θ,φ} L
    end for
    r̃ ← log D − log(1 − D) = f − log π_ω                            // reward signal
    Update ω via PPO on max_{π_ω} E[Σ_t β^t r̃]                      // policy update
    if D ≈ 1/2 on held-out data then
        break
    end if
end for
return g_θ    (disentangled, transferable reward)
```

Fu et al. (2018) use PPO (Schulman et al., 2017) with $J = 1$ discriminator step per policy step. The reward signal $\tilde{r} = f - \log \pi_\omega$ is the log-odds of the discriminator. At convergence $D = 1/2$ and $g_\theta = r^* + c$ by Theorem 1.

5 Results from the Paper

Transfer experiment (Table 1 in paper). AIRL is trained on a PointMaze where the agent navigates to a goal in a room with one wall layout. The recovered reward g_θ is then used to train a new policy in a maze with different walls. AIRL transfers successfully: the agent reaches the goal in the new maze because g_θ captures the goal location, not the path around specific walls. GAIL and MaxEnt IRL fail transfer because their recovered rewards encode the original wall layout through the entangled shaping potential. This is the paper's central empirical result.

Continuous control (Table 2). On Ant, HalfCheetah, and Hopper from OpenAI Gym, AIRL achieves cumulative reward within 5% of the expert on all three tasks. GAIL matches this performance but does not recover a reward function. MaxEntIRL underperforms on the higher-dimensional tasks (Ant) because its linear reward parameterization is too restrictive for continuous control.

State-only vs. state-action (Section 5.2). When the true reward is state-only, $g_\theta(s)$ recovers it up to a constant as Theorem 1 predicts. When the true reward depends on actions, the state-only $g_\theta(s)$ absorbs the action-dependent component into the shaping potential h_ϕ. The reward is still useful for the original environment but does not transfer cleanly because the action-dependent piece is entangled with the dynamics. This motivates the AAIRL extension by Lee et al. (2026), which handles action-dependent rewards via anchor constraints.

6 Simulation

We replicate three experiments from Fu et al. (2018) on tabular gridworlds, testing the claims that matter most for structural estimation. Run `python airl_run.py` to regenerate Table 2.

Experiment 1: Transfer across dynamics. AIRL is trained on a 7×7 gridworld where a wall blocks the direct path to the goal. The recovered reward g_θ is then used to compute a new policy in a second gridworld with the wall in a different position. AIRL should transfer successfully because g_θ captures the goal location, not the path around the wall. MCE-IRL is trained on the same data as a baseline. Its recovered reward entangles the goal location with the original wall layout through the shaping potential, so the transferred policy fails to reach the goal in the new maze.

Experiment 2: State-only vs. state-action reward. On a 5×5 gridworld with state-only reward ($r^*(s) = 1$ at goal, 0 elsewhere), AIRL with state-only $g_\theta(s)$ should recover the reward up to a constant (Theorem 1). We then repeat the experiment with an action-dependent reward ($r^*(s, a_{\text{north}}) = 0.5$ bonus) and show that the state-only g_θ absorbs the action-dependent component into h_ϕ, reducing transfer accuracy.

Experiment 3: Estimator comparison. On the 5×5 gridworld, we compare AIRL against MCE-IRL and NFXP on cosine similarity (reward direction), policy match rate (greedy action agreement), and wall-clock time.

Metric	True	AIRL
Cosine similarity	– – –	0.165
Correlation	– – –	−0.362
Policy match rate	– – –	0.600
Goal reward	1.00	43.611
Mean non-goal reward	0.00	53.019

Table 2: Reward recovery on state-only gridworld (Experiment 2). Auto-generated by `airl_run.py`.

The decomposability condition holds because the 20% slip probability makes all states reachable from all others. AIRL uses conservative policy iteration ($\alpha_p = 0.1$) and weight decay to stabilize the adversarial loop in the tabular setting. The econirl implementation uses `AIRLEstimator` with `AIRLConfig(reward_type="tabular", policy_step_size=0.1, use_shaping=True)`. Cosine similarity measures reward direction (ignoring scale). Policy match rate measures the fraction of states where the greedy action under the recovered reward agrees with the expert.

What the experiments demonstrate. Experiment 1 is the paper's central claim: AIRL's structured discriminator (3) produces a reward that transfers, while methods that entangle reward with value do not. The transfer test is the definitive check for disentanglement because it evaluates the recovered reward under dynamics the estimator never saw. If g_θ has absorbed value function artifacts, it will guide the agent along the original path (around the old wall) rather than toward the goal.

Experiment 2 tests the scope of Theorem 1. The state-only case is the theorem's intended setting and should show near-perfect recovery up to a constant. The state-action case deliberately violates the theorem's assumptions and should show degraded transfer, quantifying exactly how

much performance is lost when the state-only restriction is violated. This gap motivates the anchor constraints in the AAIRL primer.

Experiment 3 puts AIRL in context against structural estimators. On the tabular gridworld where transitions are known and the state space is small, NFXP should dominate because it solves the Bellman fixed point exactly. MCE-IRL should also perform well via feature matching. AIRL's adversarial training is slower and less stable on tabular problems, but the experiment shows whether it converges to the same answer given enough rounds. The practical value of AIRL is in continuous-state settings (Ant, HalfCheetah) where NFXP and MCE-IRL cannot operate because the state space is too large for tabular methods.

References

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